Mergin Maps Admin Essentials

A Tutorial

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Table of contents

Creating a Project in QGIS

In this tutorial you will create a new project for surveying buildings in your neighbourhood using QGIS.

1. Before we start

Please ensure you have already:

- Installed QGIS
- Signed up to Mergin Maps
- Installed the Mergin Maps QGIS plugin

2. Create a minimal project

In this section, we'll create a mimimal project that we'll further extend later in this tutorial.

1. Open QGIS

2. Locate the Mergin Maps QGIS plugin toolbar in the upper navigation panel in QGIS. Click on the **Create Mergin Maps Project** button

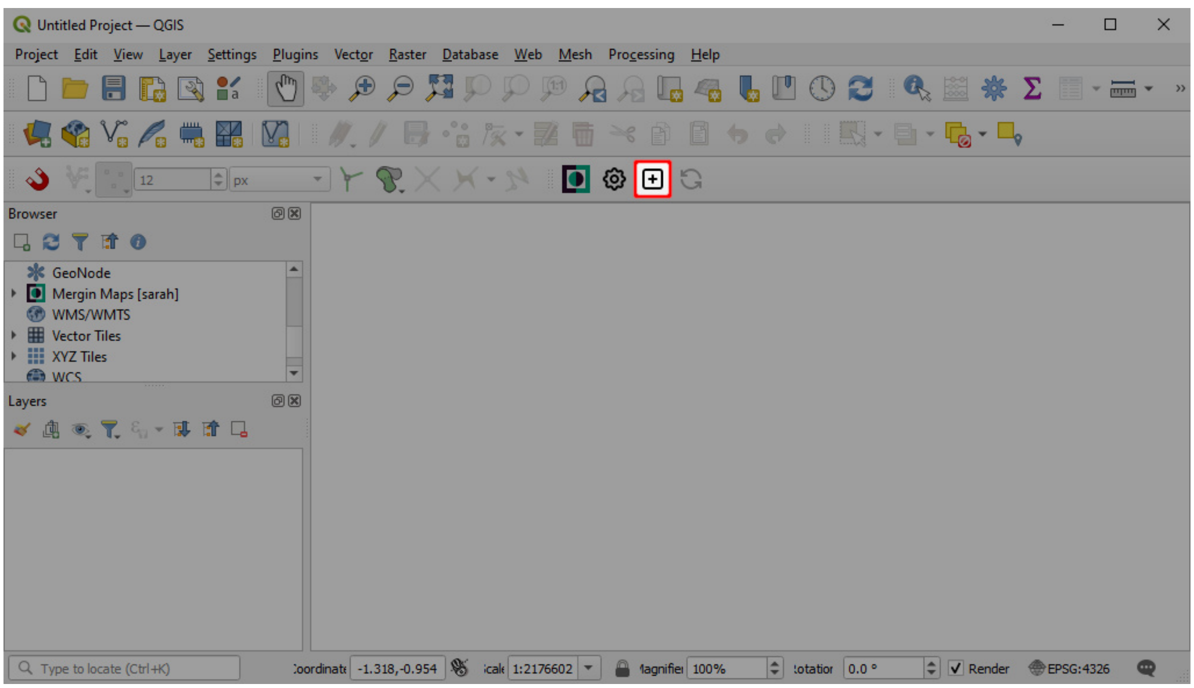

Figure 1: QGIS plugin Create Mergin Maps Project

3. Choose the first option: **New basic QGIS project**

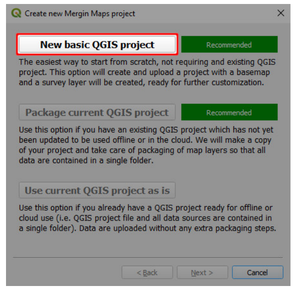

Figure 2: Create Mergin Maps project: New basic QGIS project

4. When creating a Mergin Maps project, we need to set:

- A workspace. If you have multiple workspaces available, choose the one you want to use. Here we'll use `MM Admin Essentials`.
- **Project Name**. In this example we use `buildings`. For this training you should give a unique name to the project, because all participants work in the same workspace. Therefore, add your name with and underscore (for example `buildings_hans`)
- The folder where the project should be created. Here, we use `MerginMapsProjects`

5. Click **Finish** to create the project.

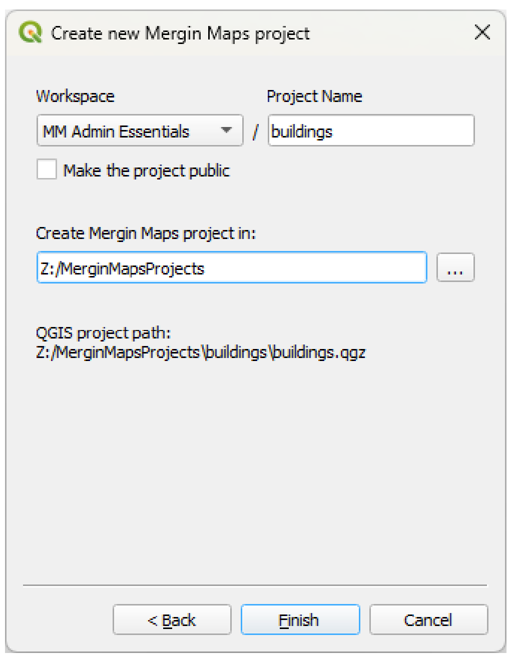

Figure 3: Create new Mergin Maps project in QGIS

Your new project should now be created and uploaded to Mergin Maps cloud.

6. **Close** the dialog window to continue working with this project in QGIS.

3. Create a subfolder for photos

It can be useful to set up a custom folder for photos, e.g. if you want to use selective synchronisation or if you simply want to have your data organised. In this tutorial, we'll use a subfolder for selective sync of our photos.

To set up a custom folder:

1. In the **Browser** panel, click right on **Project Home** and choose **Open Directory...** from the context menu

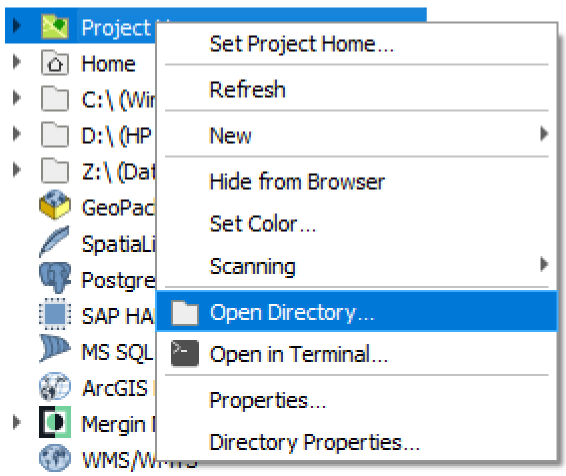

Figure 4: Open Directory

2. There create a subfolder with the name `photos`

Later, we'll set up this folder for storing our photos and the use of selective synchronisation.

4. Add layers

This basic project contains a single point layer called `Survey` and an OpenStreetMap vector tiles layer as a background map.

Various background maps can be used in Mergin Maps mobile app, if they are supported by QGIS. You can learn how to add offline and online background maps here.

We will now add two more survey layers - a point layer for surveying buildings and a line layer for road or path close to the building.

1. Select **Layer > Create Layer > New GeoPackage Layer...**

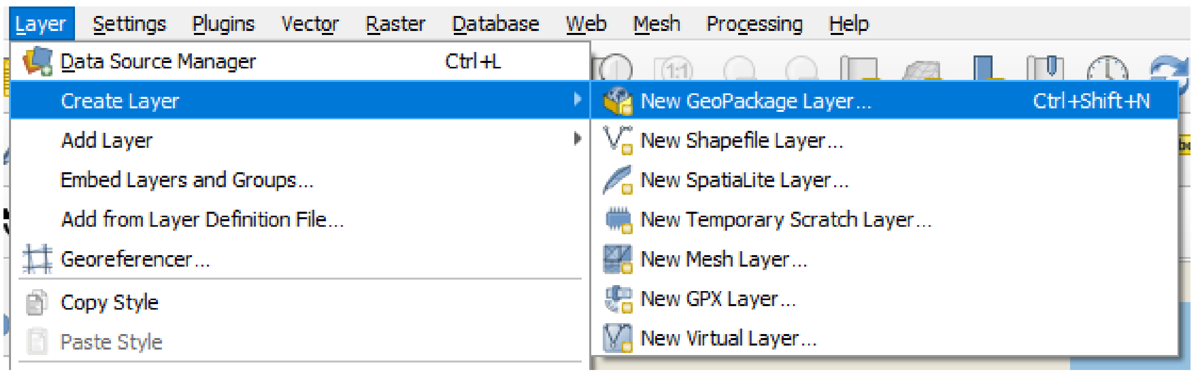

Figure 5: QGIS Create new GeoPackage layer

2. Now we will create the `buildings` layer:

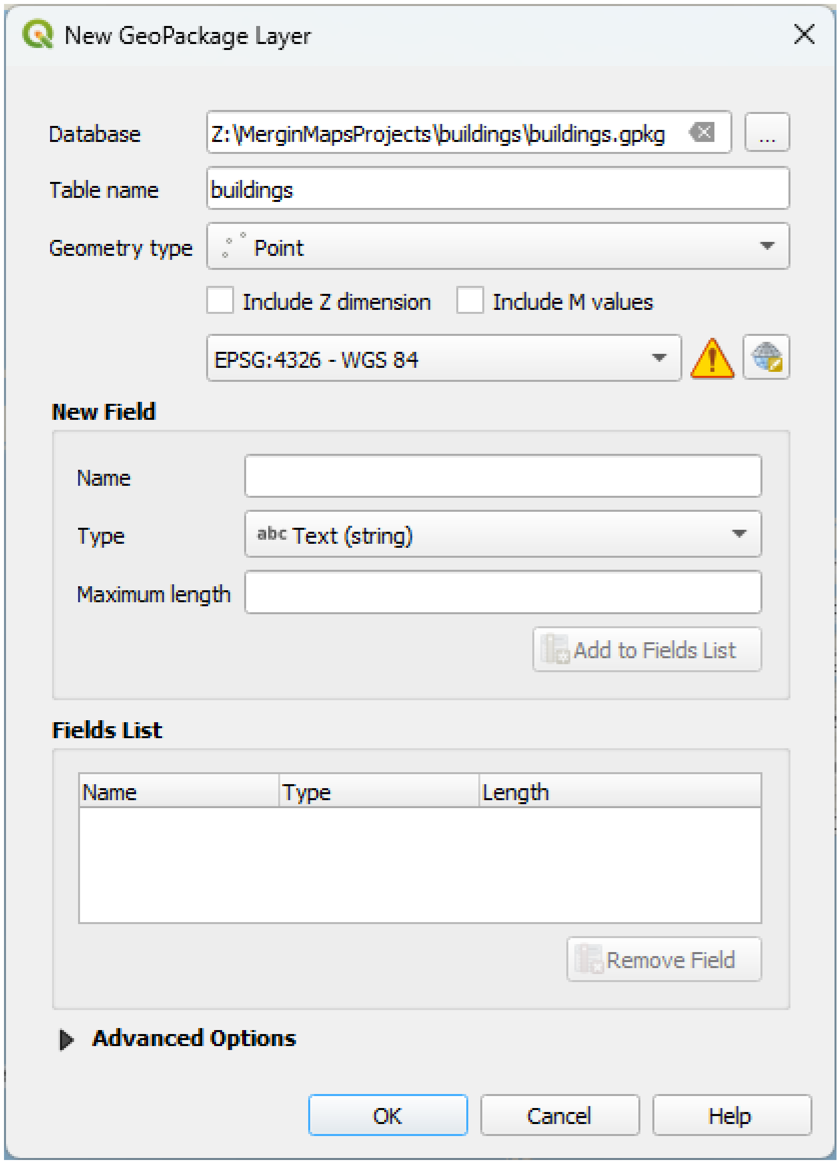

Figure 6: New GeoPackage point layer

- **Database**: navigate to the project's folder and save the GeoPackage as `buildings.gpkg` Here: `MerginMapsProjects\buildings\buildings.gpkg`
- **Table name** will be set as `buildings` by default
- **Geometry type**: ***Point***

3. Add a **New Field** called `date` with the data type ***Date***

Figure 7: New date field

4. Add three more new fields:

   - `floors`, data type ***Integer (32 Bit)***
   - `function`, data type ***Text (string)***
   - `photo`, data type ***Text Data***

   In the **Fields List**, you can see the overview of added attributes and their respective data types. It should look like this:

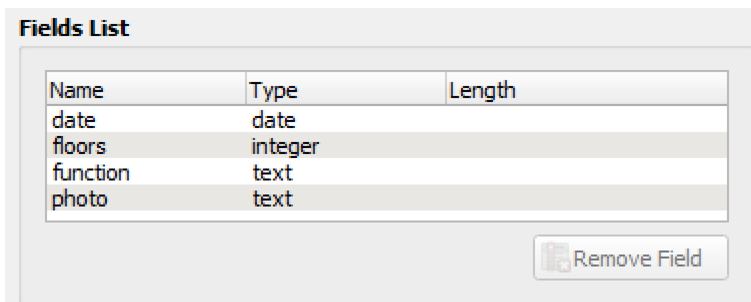

Figure 8: Field list with created attributes

5. Click **OK**. A new layer called `buildings` has now been added to the **Layers** panel.

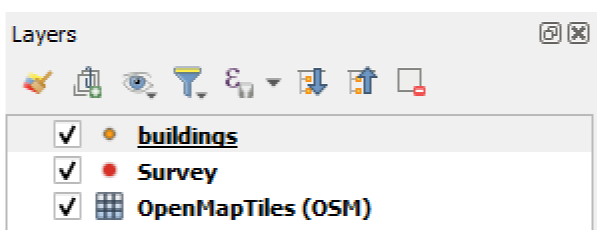

Figure 9: Point layer added to QGIS project

6. Repeat steps 1 to 5 above to add another new layer called `roads` with these details:

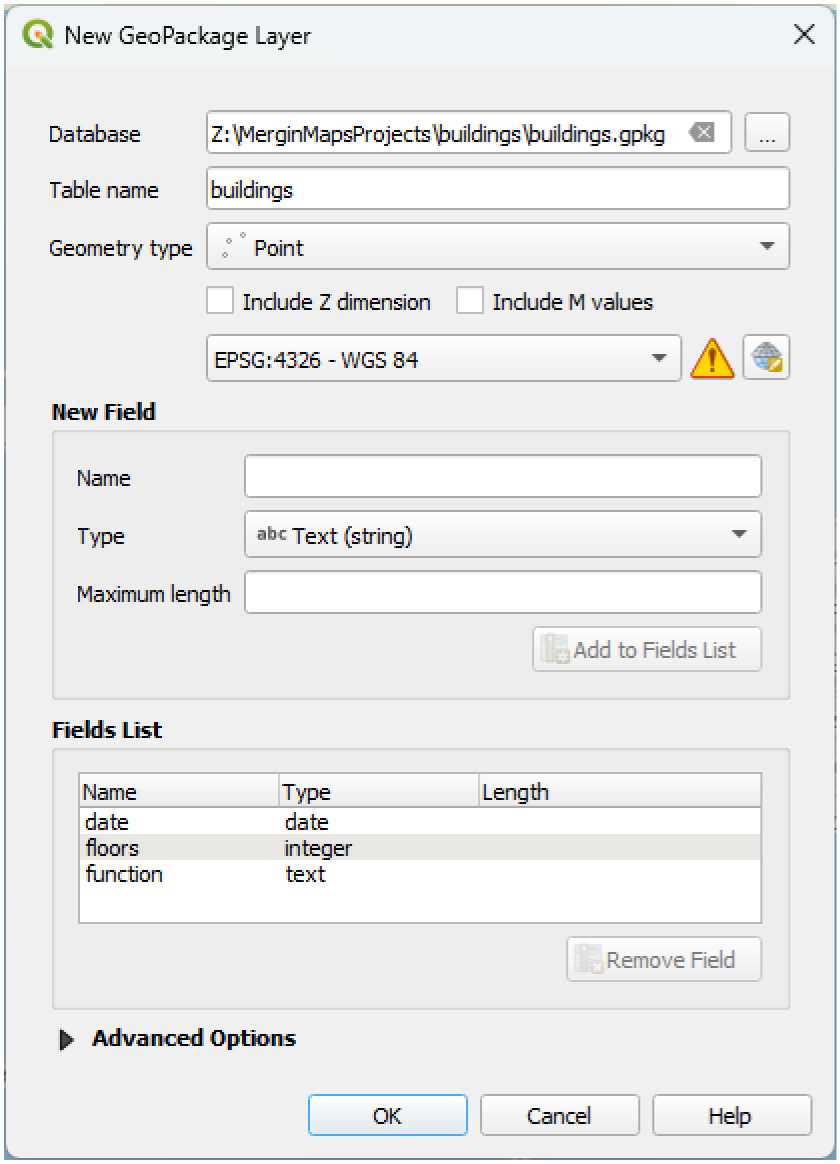

Figure 10: Create GeoPackage line layer with defined attributes

- **Database**: `MerginMapsProjects\buildings\roads.gpkg`
- **Geometry type**: ***LineString***
- **Fields**:
  - `type`, data type ***Text (string)***
  - `sidewalk`, data type ***Boolean***

Once you are finished, there should be two new layers on the **Layers** panel: `buildings` and `roads`.

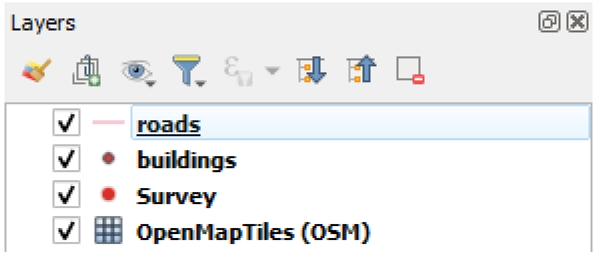

Figure 11: Layers panel in QGIS with created layers

5. Configure attributes forms

Now we will make a couple of small changes to the layers' form settings. The forms define how users interact with feature attributes in the field.

We will start with setting up the form for the `buildings` layer:

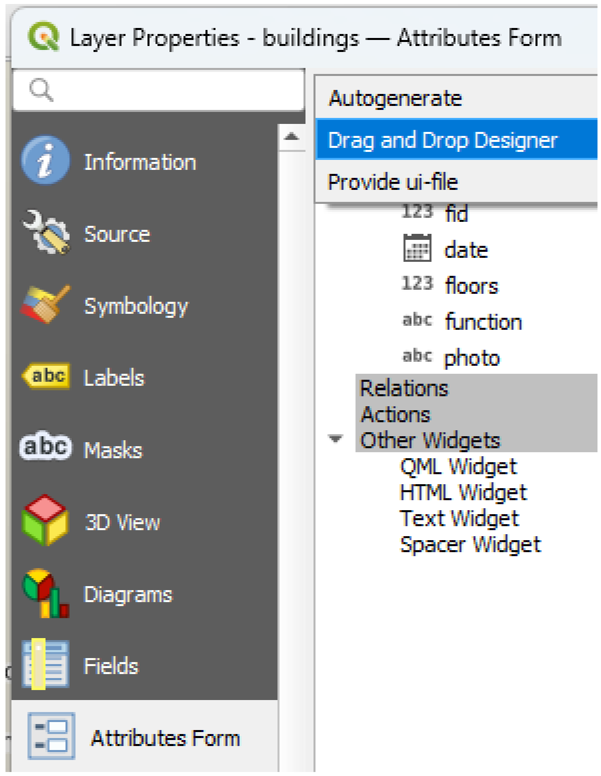

Figure 12: QGIS Attributes form

1. Double-click the `buildings` layer in the **Layers** panel to open the **Layer properties**
2. Select **Attributes Form** on the left change ***Autogenerate*** to ***Drag and Drop Designer***. This will allow us to easily modify the order of attributes in the field form and remove the ones that we don't want to show.
3. Remove the `fid` attribute by clicking the minus button.

   `fid` is a special attribute used to identify features. We recommend not allowing users to edit this attribute.
4. Drag the `photo` attribute to the top.

   The screen now looks like this:

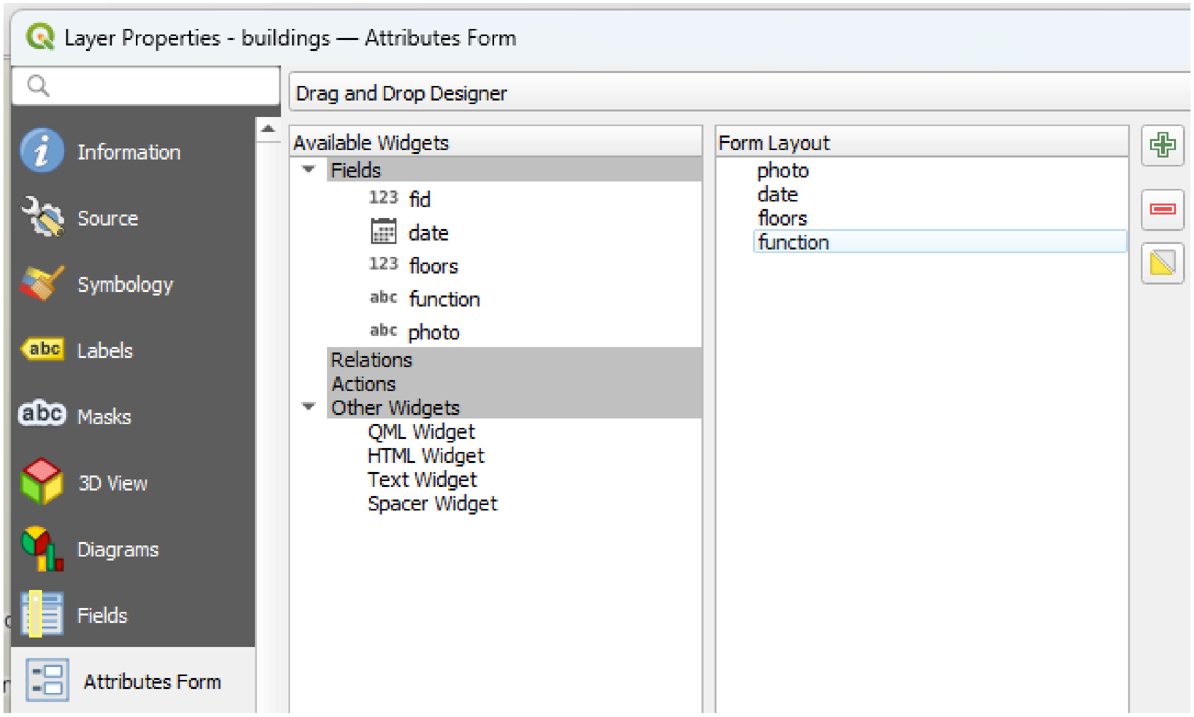

Figure 13: QGIS Attributes form modified

The `photo` attribute itself is stored as ***Text Data*** and we will use it to attach photos to surveyed features. To achieve this, we need to configure the `photo` attribute's **Widget Type** as ***Attachment***.

5. Click on the `photo` attribute
6. At **Alias** type `Photo`
7. Change **Widget Type** to ***Attachment***

   Path defines where the photos will be stored. We need to change the **Default path** to the `photos` folder we have created earlier.

8. Click on the *Data defined override* icon and choose **Edit...**

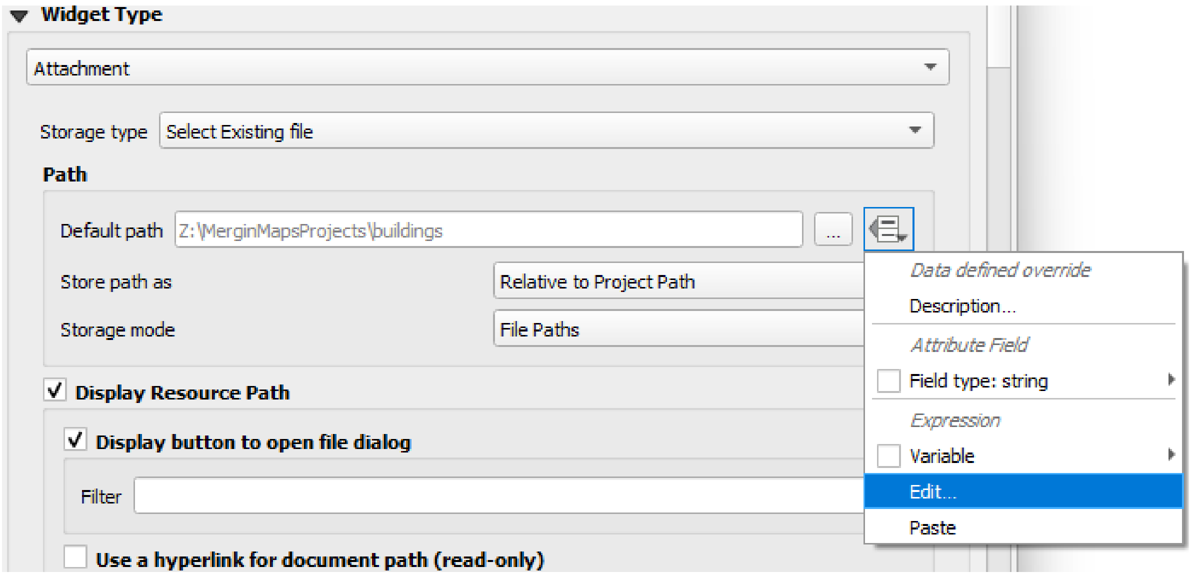

Figure 14: Data defined override

9. In **Expression String Builder** enter the expression `@project_folder + '/photos'`. Click **OK**

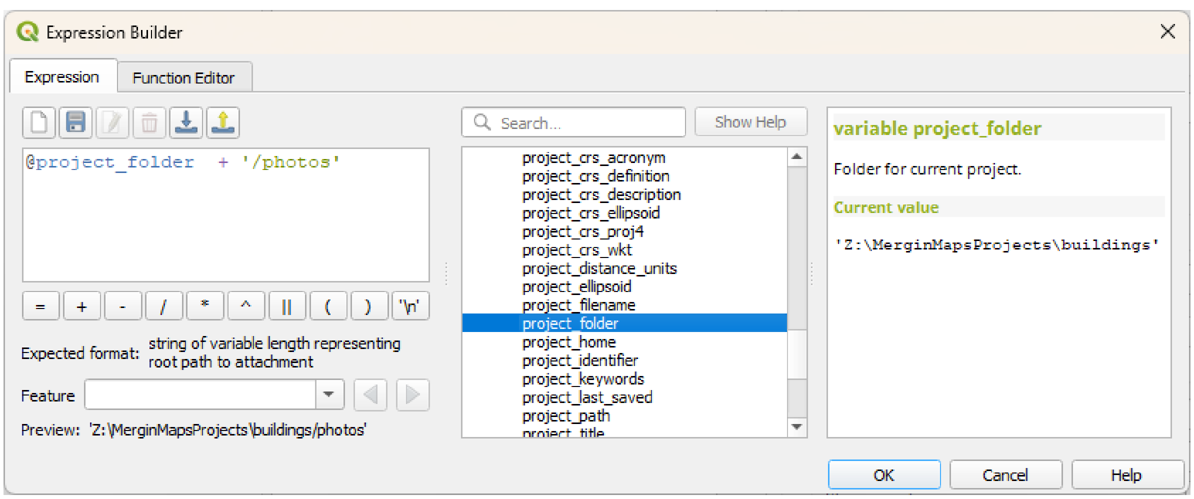

Figure 15: Expression for path

10. Set *Store path as* **Relative to Project Path**:

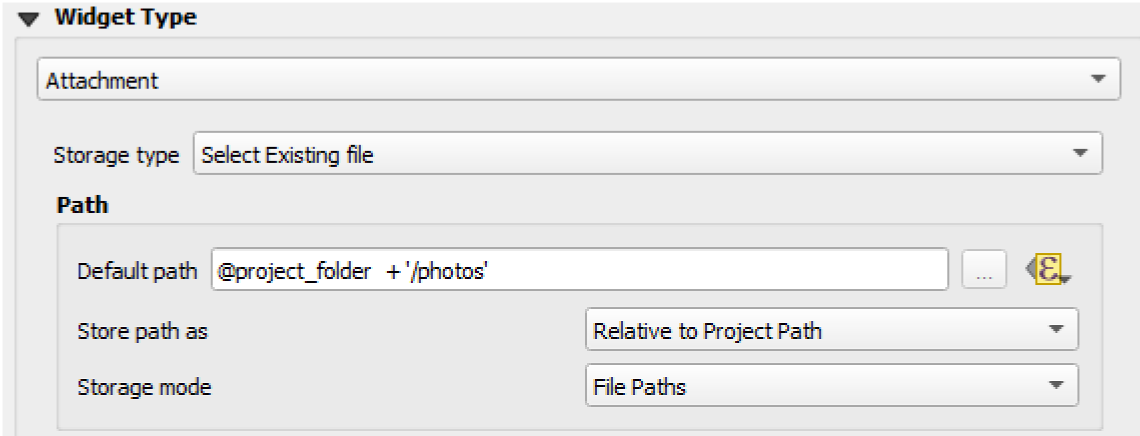

Figure 16: Relative path

11. Set **Integrated Document Viewer** type to ***Image***.

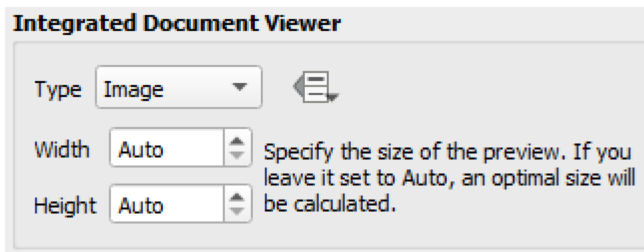

Figure 17: Integrated document viewer

Now let's modify the `date` attribute, so it shows the date in the desired format.

12. Click on the `date` attribute
13. Set **Alias** to `Date`
14. Under **Widget Type** use **Display Format** ***Custom*** and change it to `dd MMMM yyyy`. Check the preview.

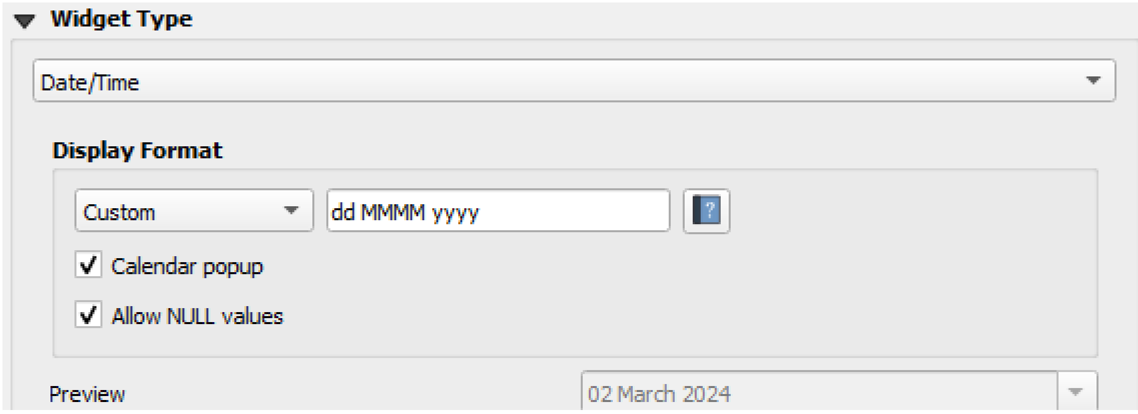

Figure 18: Date widget

It's useful to have the current date automatically filled in when you add a point in the Mergin Maps mobile app.

15. Under **Defaults** use `$now` as **Default value**. Check the preview.

    Now we'll configure the `floors` attribute form field.

16. Set the form for the `floors` attribute as follows:

    - **Alias** to `Floors`
    - **Widget Type** to ***Range*** using a ***Slider***. Set **Minimum** to `1` and **Maximum** to `30`. Feel free to adjust the maximum to your needs. We'll use `1` for **Step**.

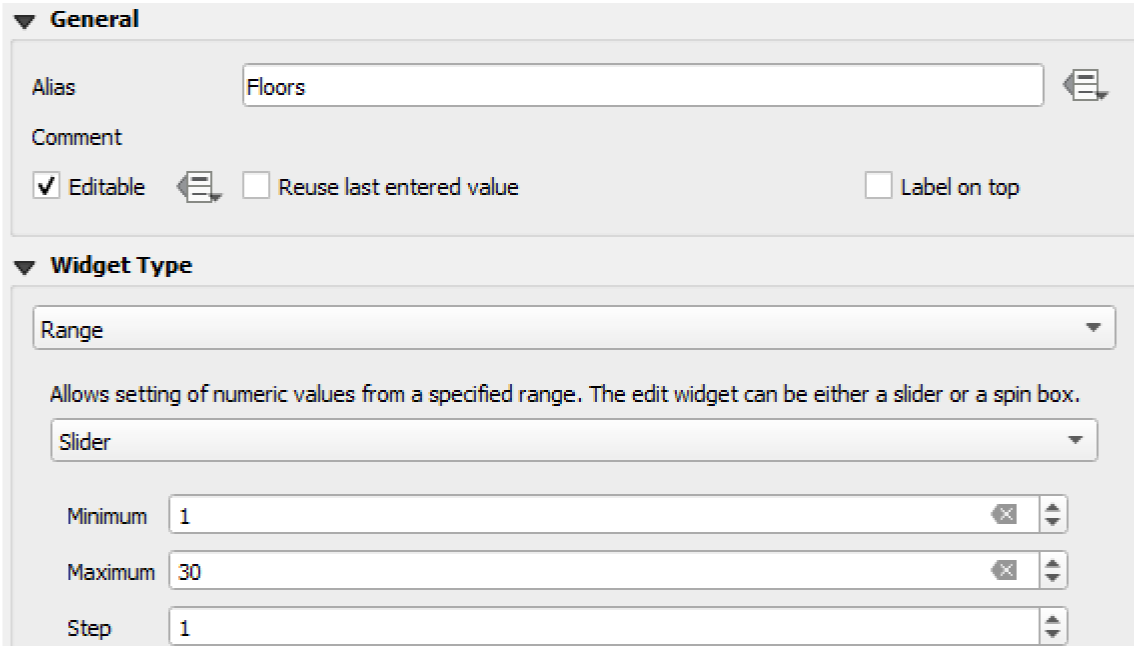

Figure 19: Slider widget

Finally, we'll modify the `function` attribute for the function of the building. We'll use a drop-down list.

17. Set the form for the `function` attribute as follows:

    - **Alias** to `Function`
    - Change the **Widget Type** to ***Value Map***
    - Enter **Values** and **Descriptions** similar to these (both are set the same in this example):

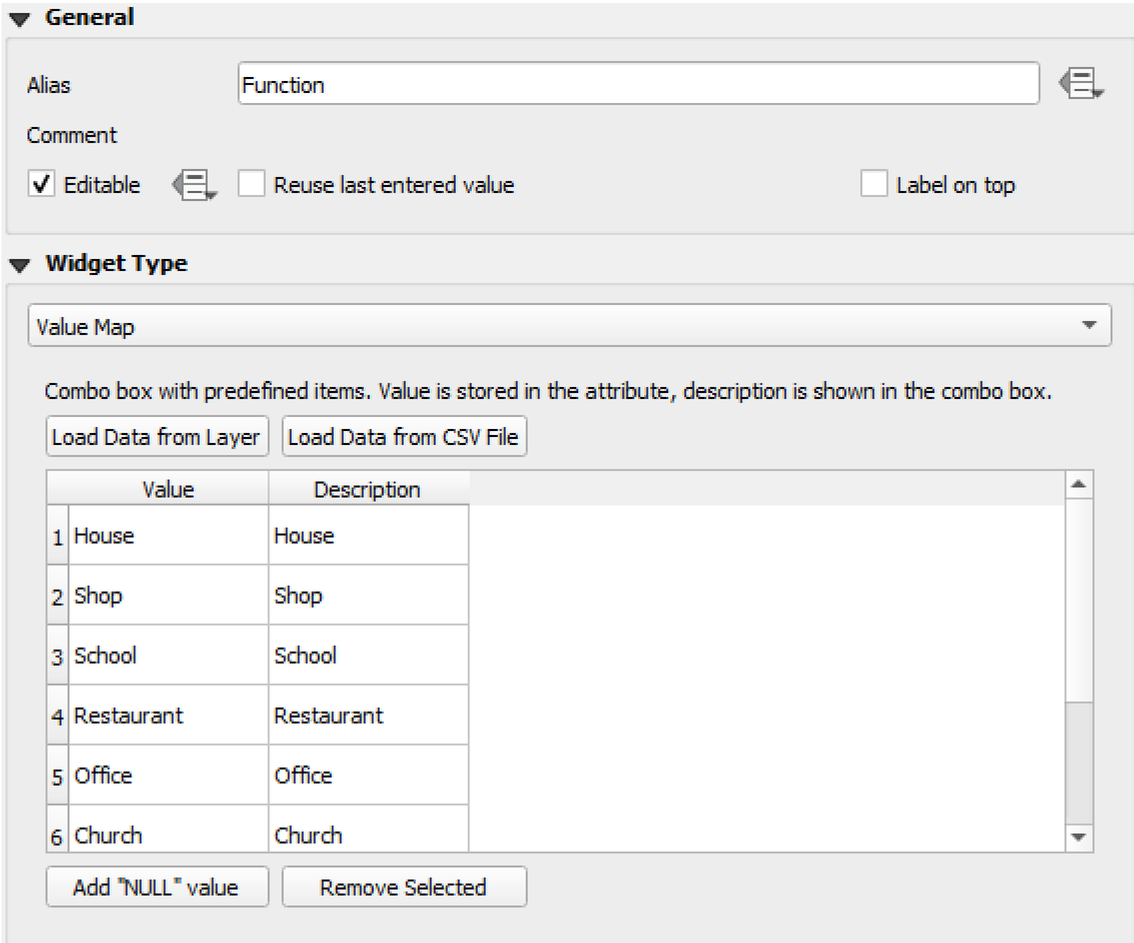

Figure 20: QGIS Attributes form Value Map Function

Value is how the data will be stored in the underlying dataset and **Description** is how it will be displayed in the form to the user.

18. The setup of the `buildings` layer is complete. Click **OK** to close the **Layer Properties** window.

    Let's configure the `roads` layer now. This time we'll not use the ***Drag and Drop Designer***.

19. In the **Layers** panel, double-click the `roads` layer to open the **Layer properties**

20. Select **Attributes Form** on the left and click on the `type` attribute

21. Set the form for the `type` attribute as follows:

    - **Alias** to `Road type`

- **Widget type** to ***Value Map***
- Enter **Values** and **Descriptions** similar to these (both are set the same in this example):

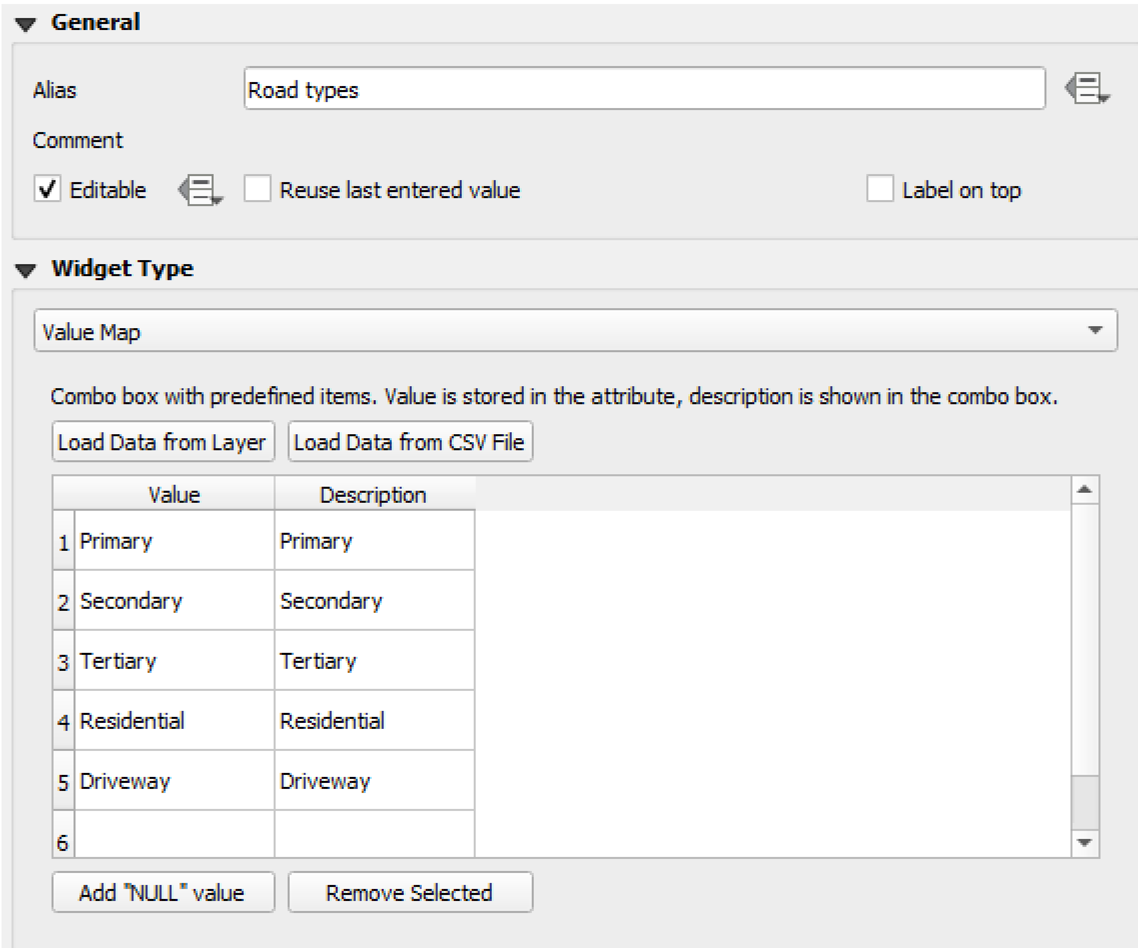

Figure 21: QGIS Attributes form Value Map Types

22. Click on the `sidewalk` attribute.

    Because the data type of the `sidewalk` field is ***boolean***, by default the ***Checkbox*** widget type is used and ***"True" or "False"*** are stored. We only need to set the **Alias**.

23. Set the **Alias** to `Is there a sidewalk?`
24. `fid` should not be editable. Click on the `fid` attribute and uncheck the **Editable** option.
25. Click **OK** to close the **Layer properties**.

There is much more that can be done with the Attributes Form in QGIS when preparing a Mergin Maps project! You can learn more about forms in Setting Up Form Widgets, Advanced Form Configuration or this video.

6. Project settings

In this section, we'll configure the project properties.

Relative paths

All paths to the project data in the Mergin Maps mobile app are relative to the project location. Let's check if this is the case in the **Project Properties**.

1. In the main menu, go to **Project** > **Properties**

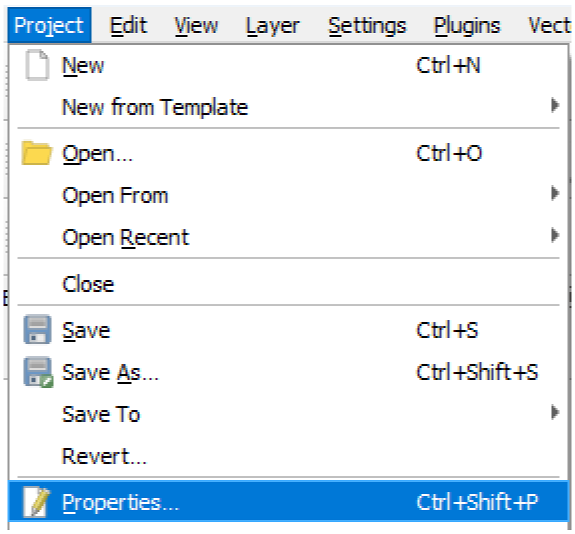

Figure 22: Project properties

2. Ensure that the paths are set to *Relative* in the **General** tab in **Project Properties**

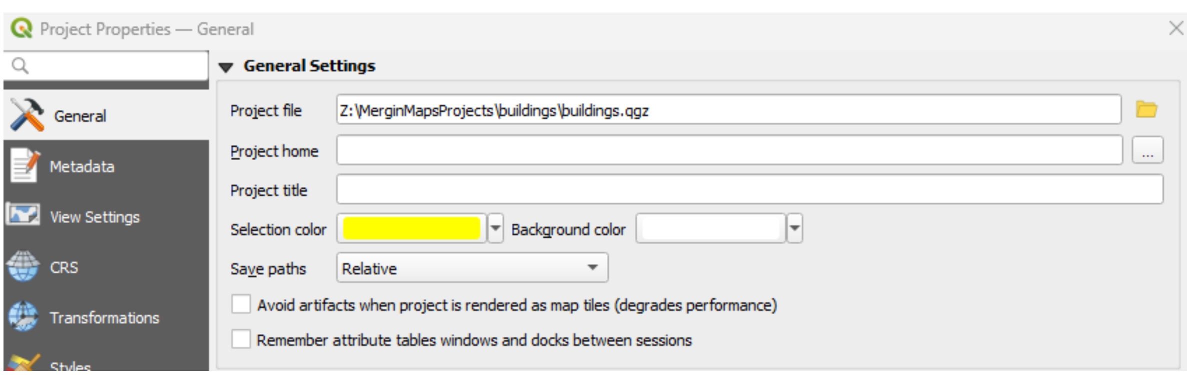

Figure 23: Project properties relative paths

Layer capabilities

Next, we'll check **Layers capabilities**.

3. In the **Project Properties** dialog, go to the **Data Sources** tab.

    - Identifiable layers can be queried in Mergin Maps mobile app. If you want to be able to search for attribute values in a layer, it needs to be **identifiable** and **searchable**.
    - **read-only** layers cannot be modified. If a vector layer is not intended to be used as a survey layer, set it as read-only.
    - In case you have non-spatial layers, they need to be set as **searchable** to enable browsing, searching, or editing.

4. Check the settings.

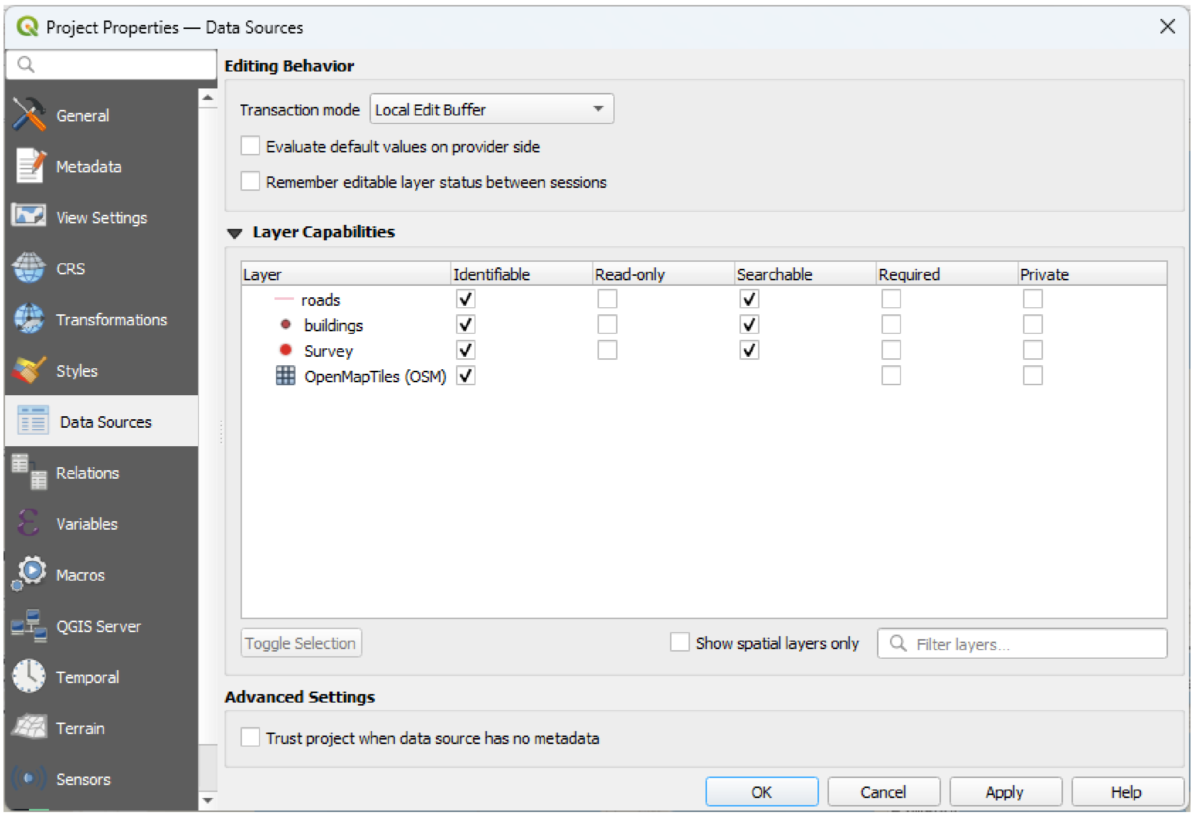

Figure 24: Data sources

Project extent

In the Mergin Maps mobile app, there is an option to zoom to the project extent.

If the project extent is not set, the Mergin Maps mobile app zooms to all visible layers. This is not particularly convenient when you have a layer with a large/global extent (e.g. OpenStreetMap).

To set the project extent:

5. In the **Project Properties** dialog, go to the **View Settings** tab
6. Check the **Set Project Full Extent** option

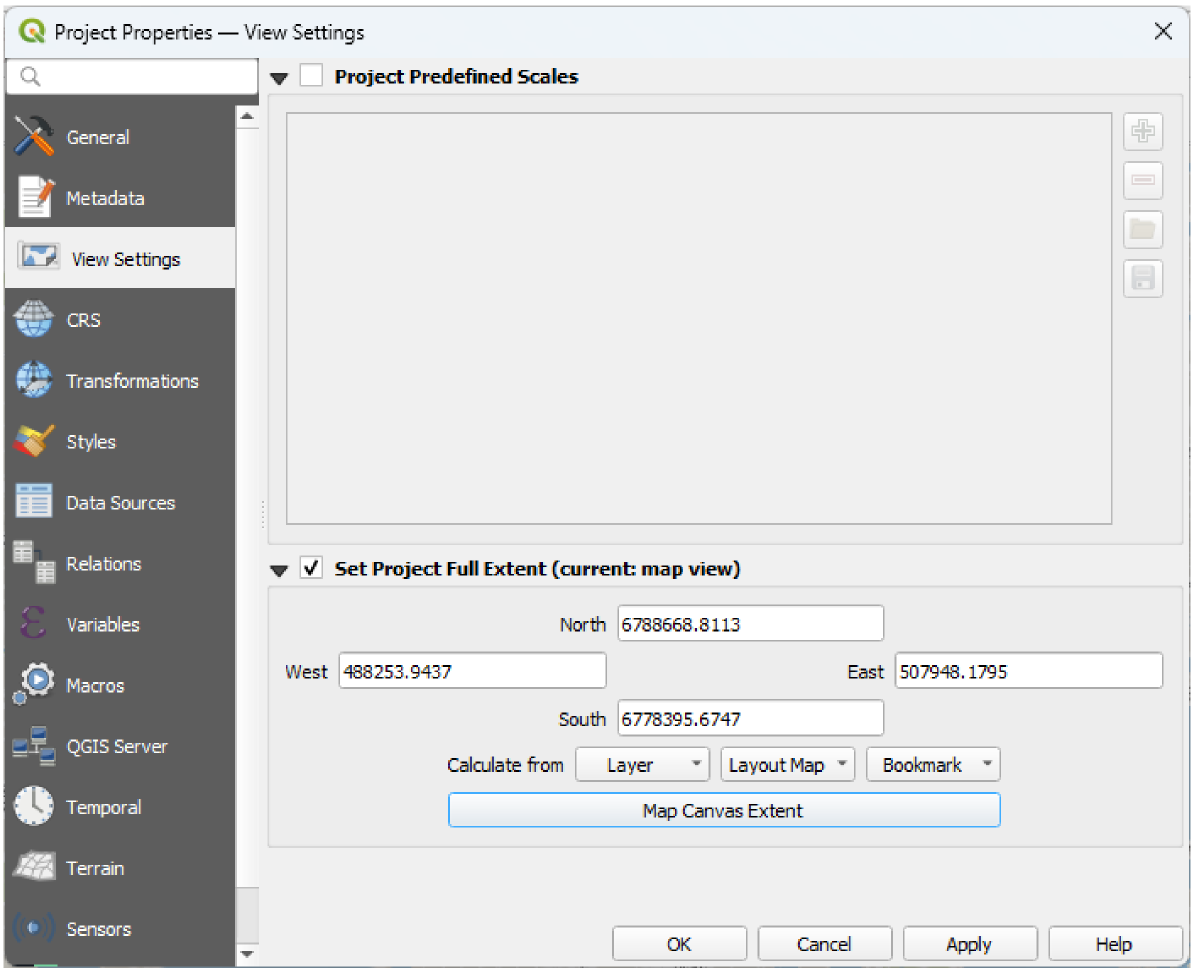

Figure 25: Project extent

Here, either enter the coordinate extent of your project bounding box or use the map canvas extent. The extent can be also calculated from a layer in your project.

7. Set the extent to your neighbourhood

Selective synchronisation

Selective sync feature adds a possibility to not download specified files on other devices in the synchronisation process. These files are only stored on the creator's device and server and can be accessed on Mergin Maps web or QGIS desktop. Other collaborators on different devices will not receive these files during synchronisation.

Selective sync is useful mainly when a project contains a lot of data (for example photos) and these data do not necessarily need to be stored on all devices. Another advantage is a significant reduction of synchronisation time.

See the example in the picture below. Two surveyors Jim and Susan are capturing features in the field. When it comes to synchronisation, Jim hits the arrow icon to synchronise his changes. Features together with photos are now stored on the server. When Susan synchronises the project, synchronisation first downloads Jim's changes (including photos) and after that uploads Susan's changes to the server. However, selective sync can exclude photos from being downloaded.

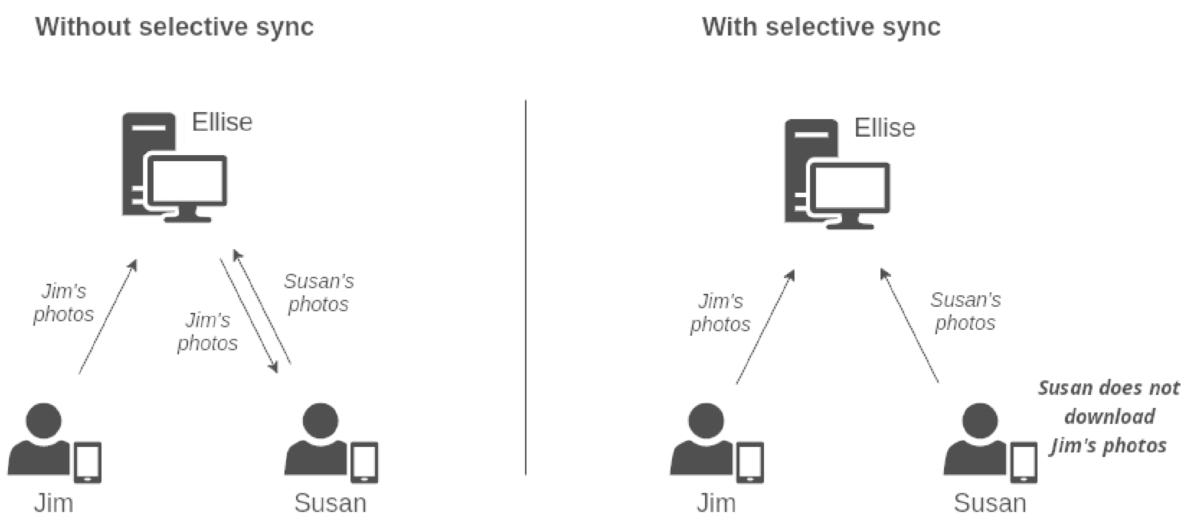

Figure 26: Selective sync example

Let's set it up for our *photos* folder.

8. In the **Project Properties** dialog, go to the **Mergin Maps** tab
9. Check **Enable selective sync for Input**
10. At **Only apply for folder** type `photos`, so it only applies to this subfolder of your project

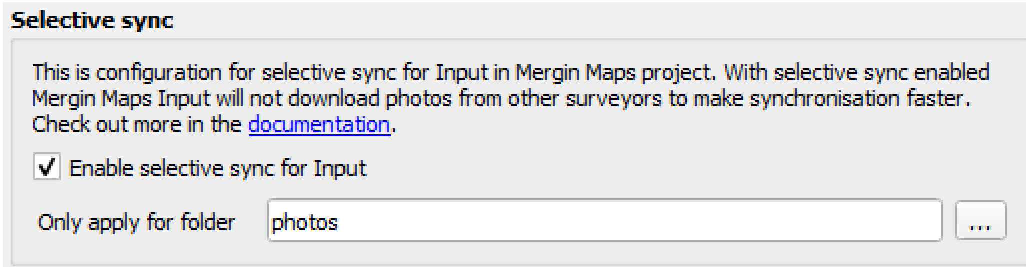

Figure 27: Selective sync for a subfolder

Resizing pictures automatically

Photos that are captured during the field survey or uploaded using Mergin Maps mobile app can be automatically resized, e.g. to save up storage space.

By default, the quality is set to Original - the original pictures are stored. If you want to resize the pictures, you can choose from High, Medium, or Low quality. The EXIF metadata of the original files are kept.

11. Set the **Photo quality** to ***Medium***.

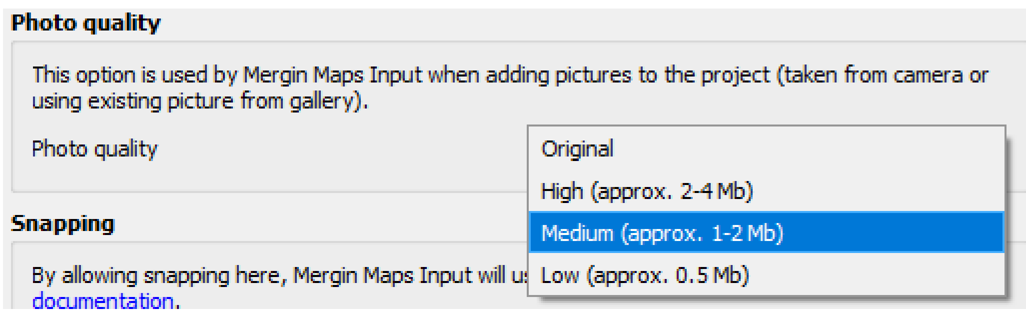

Figure 28: Photo quality

Snapping

For the roads it would be useful to switch on snapping. You can snap the vertices of new features to existing geometries. Snapping can also help you avoid creating topological errors in your datasets.

The Mergin Maps QGIS plugin provides three snapping options:

- *No snapping* - snapping is not enabled (default)

- *Basic snapping* - features are snapped to the vertices and segments of vector features in the project
- *Follow QGIS snapping* - uses the snapping preferences defined in teh QGIS project

Here we'll setup *Basic snapping*:

12. Change the snapping settings to ***Basic snapping***

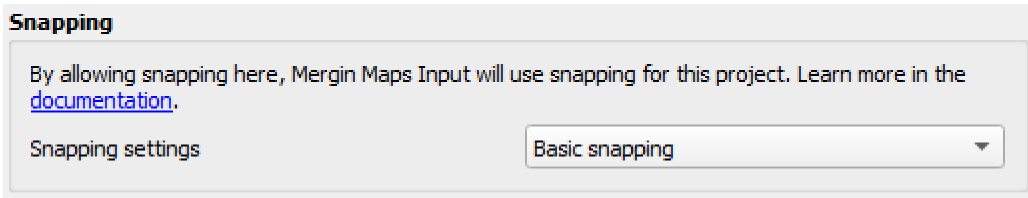

Figure 29: Basic snapping

13. Click OK to close the dialog

When capturing a new feature near an existing one, the crosshairs will turn purple and snap to its vertex or to its segment.

For more advanced snapping settings, check the Mergin Maps documentation.

Map sketching

Map sketching lets you draw freehand sketches on the map using a line layer. We'll enable this in our project to see how the feature works:

1.Tick the box to enable map sketching

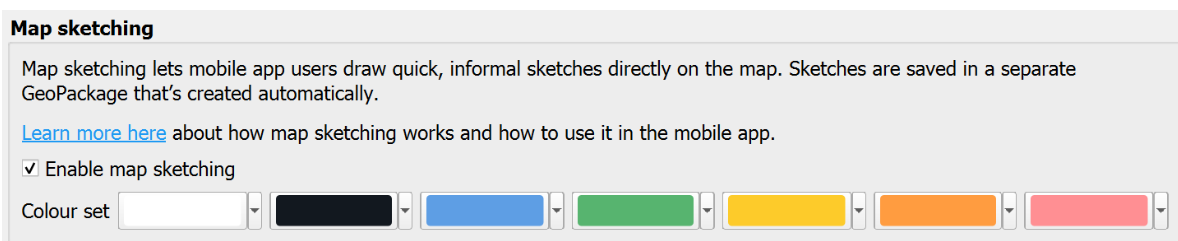

Figure 30: Map sketching

2. *Optional* You can use the drop-down boxes to select custom colours for your sketching layers or use the default options

Other settings

In the **Project Properties** under the **Mergin Maps** tab you can configure a few more things:

- Photo name format: you can use expressions with variables to control the naming of photos
- Position tracking: when this is activated, you can track your position with the Mergin Maps mobile app
- Photo sketching: similar to map sketching, this enables drawing freehand notes on photos taken in the app, this feature is still experimental so may cause unexpected behaviour
- Layer order: choose how layers are ordered when viewed in the 'Layers' tab, by default this will follow layer order in QGIS but you can set it to 'Alphabetical' if you prefer

For this tutorial, we'll not change these settings. Check the Mergin Maps documentation if you want to learn more about this.

7. Settings for Mergin Maps mobile app preview panel

What appears in the Mergin Maps mobile app preview panel can be defined in the **Display** tab in **Layer Properties**. Here we'll configure this for the `buildings` layer.

1. In the **Layers** panel, double-click on the `buildings` layer
2. Go to the **Display** tab

At **Display Name** you can add a field or expression. This will be shown in the title of the preview panel in the app.

3. Add the following expression:

   ```
   "function" || ' surveyed on: ' || "date"
   ```

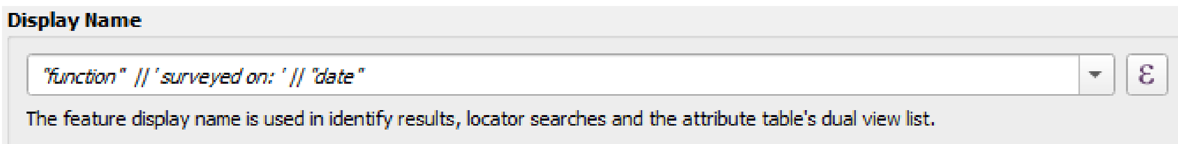

Figure 31: Display Name

In addition to this, you can also add the picture or other attributes of the survey feature using the **HTML Map Tip**. The **HTML Map Tip** is the content of the preview panel. While QGIS always interprets the content of map tip as being HTML, Mergin Maps mobile app extends the syntax to allow two more modes: field values and images. If the map tip is not specified, Mergin Maps mobile app will try to use the first three fields and show their attribute values.

Here we're going to add a photo.

4. Add the following **HTML Map Tip**:

   ```
   <img src="file:///[%@project_folder%]/[% "photo" %]" width="300">
   ```

5. Click **OK** to close the dialog.

8. Style the layers

The layer styles from QGIS are also used in the Mergin Maps mobile app. Here we're going to embed an SVG marker.

1. In the **Layers** panel select the `buildings` layer and type `F7` to open the **Layer Styling** panel
2. There click **Simple Marker** and change the ***Symbol layer type*** to **SVG Marker**
3. In the **SVG browser** look for **accommodation** > **house**
4. Change the **Size** to 6 mm. Change the colour if needed

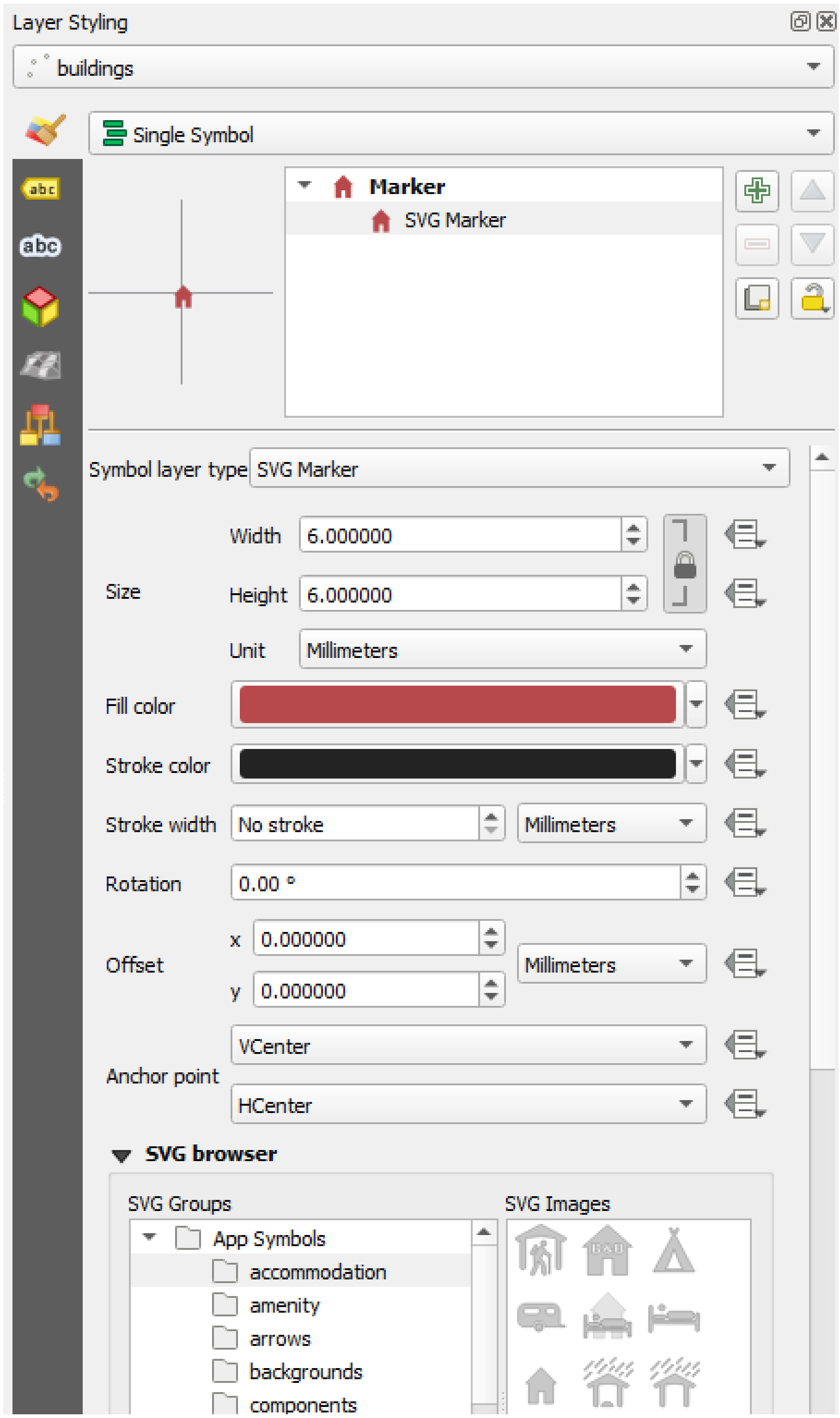

Figure 32: Select SVG symbol

5. Scroll down and under **Dynamic SVG parameters** click the arrow next to the file path and choose ***Embed File...***

6. Browse to the symbol you want to embed (e.g. `accommodation_house.svg`) and click **Open**. Now this file will be embedded in your project and used to style the `buildings` layer.

9. Saving changes to Mergin Maps

We'll now make some last changes, save them and sync the project back to the Mergin Maps cloud.

1. Right-click on the layer called `Survey` and find the **Rename Layer** option. Rename the layer to `Survey notes` (the name `Survey` is too general in this context).

2. **Save** the QGIS project

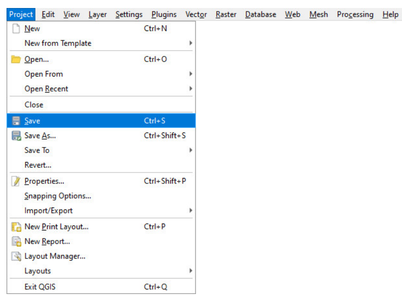

Figure 33: Save QGIS project

3. Use the **Synchronise Mergin Maps Project** button in the Mergin Maps QGIS plugin toolbar.

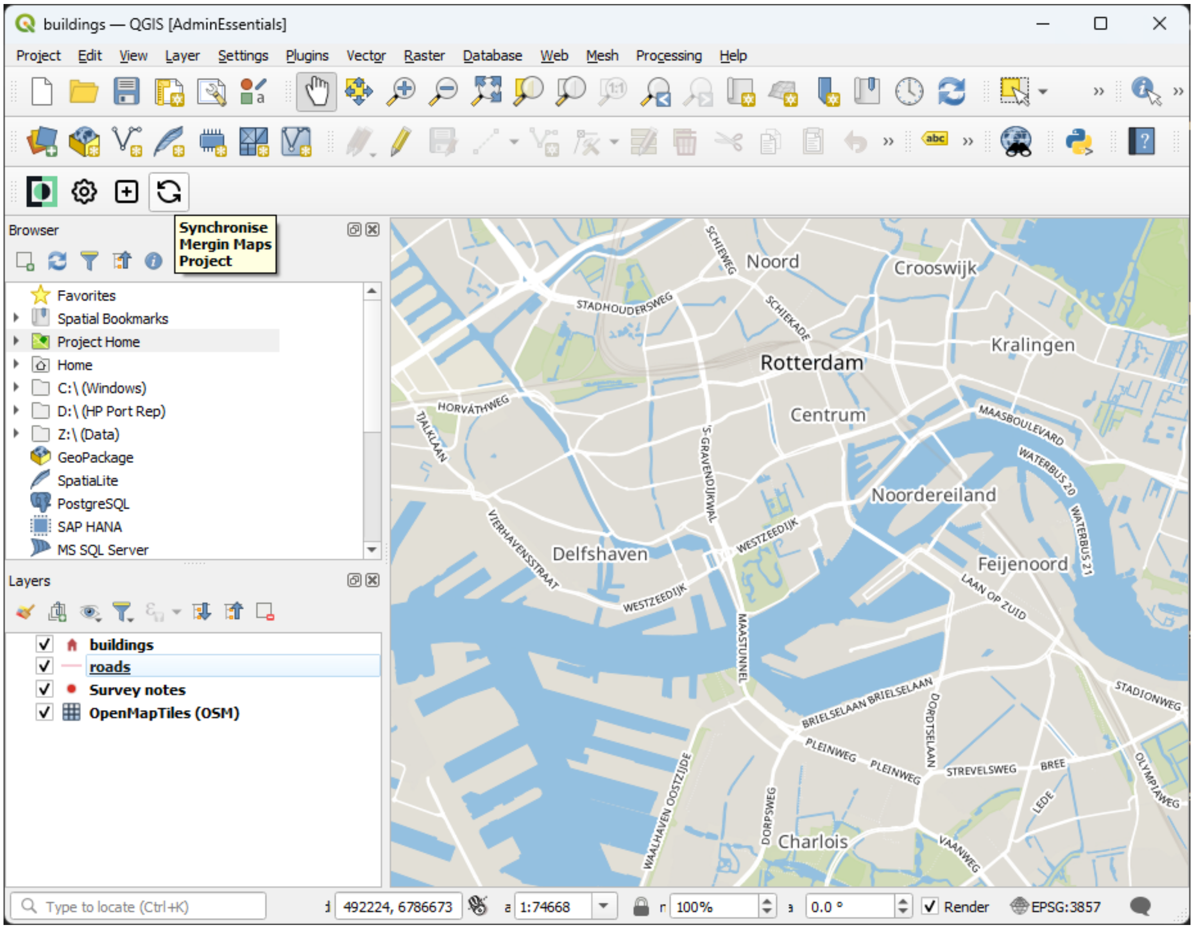

Figure 34: Synchronise Mergin Maps project in QGIS

The **Project status** window will open. It contains the overview of local changes that were made since the last synchronisation: two layers were added to the project (`roads.gpkg` and `buildings.gpkg`) and some changes were made in the QGIS project file (`buildings.qgz`).

4. Click **Sync** to synchronise the project

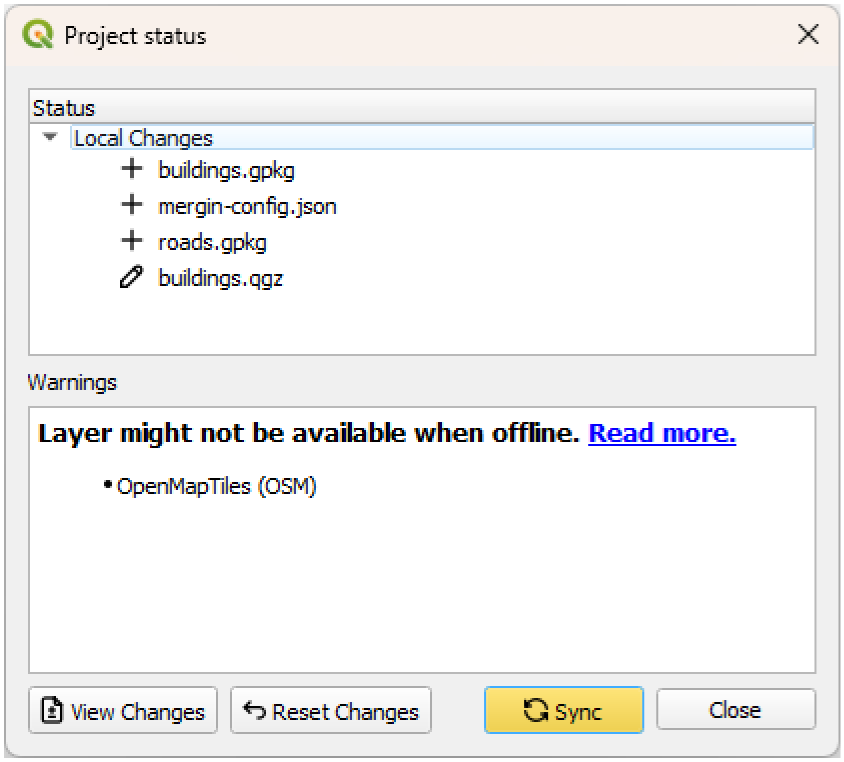

Figure 35: Mergin Maps plugin synchronisation Project status

In a few moments your changes are safely stored in the Mergin Maps cloud

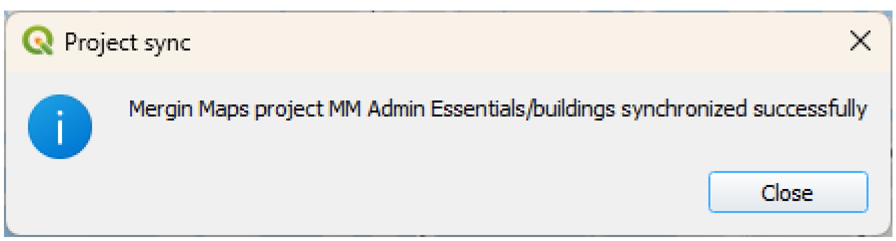

Figure 36: Successful synchronisation of Mergin Maps project in QGIS

Synchronising changes between users and devices is a core principle of Mergin Maps. When you sync a project, changes that have been made by other users and devices since you last synced are fetched and any changes you've made are pushed.

Changes are merged safely and easily from different users, even when they edit the same feature. Mergin Maps tracks project version history so you can download a previous version of a project if you need to.